Actual first-jet precision and necessary factor count

Same orientation completion, 14 November 2026
Source `8455c485411b19c3b6a99cfc6c95f32276867b9b`

Scope. A constructed precision upper control and a necessary lowest-layer factor count for the actual first crossing jet. These are unsigned source-bound claims; no signed standing is asserted [pcc:unverified].

Actual objects. Keep the parent's literal completed polynomial $W_{xy}(T)$, with retained bit mask, grid size 2^a and positive retained multiplicities m_j. At a retained pole $t = -2^i$, let J_{xy} be coefficient one of $W_{xy}(t + 2^{a+1}Z)$. The parent derives its actual nonzero diagonal $c = J_{00}$ and exact weight

$$w = v_2(c) = a + 1 - i + G_i, \qquad G_i = m_i - \sum_{i<j<u} (j - i)m_j.$$

Write $R_{xy} = J_{xy}/c$ at crossings with equal bit i, and $R_{xy} = 0$ otherwise. Assume $i < u \le a$. No arbitrary good scalar or low-rank witness is supplied. Node/origin values and source degree remain those of the same polynomial family.

Precision from actual row residues. If $u \le v \le a$ and $x \equiv z \pmod{2^v}$, their retained masks agree. For every selected anchor $q = 2^j$, its actual denominator has valuation at most $j < u$; the pole denominator has valuation at most $i < u$. Thus

$$\frac{q - \delta(z, y)}{q - \delta(x, y)} - 1 = \frac{\delta(x, y) - \delta(z, y)}{q - \delta(x, y)}, \quad v_2\left(\frac{q - \delta(z, y)}{q - \delta(x, y)} - 1\right) \ge v - u + 1.$$

Zero differences are handled by the exact predicate $\mathrm{VBound}(k, q) \equiv (q = 0 \vee v_2(q) \ge k)$. Actual polynomial first-jet identities give $R_{xy} = R_{zy}$ times a finite product of these near-one ratios. Derived prototype integrality yields

$$\mathrm{VBound}(v - u + 1, R_{xy} - R_{zy}).$$

Noncrossing entries vanish explicitly. This is a quantified source proof, not a sampled field certificate.

Constructed factors. Choose the literal valid row $z = x \bmod 2^v$. For $r \in \{0, \dots, 2^v - 1\}$, set

$$A_{xr} = [x \bmod 2^v = r], \qquad B_{ry} = R_{ry}.$$

The exact finite product is $(AB)_{xy} = R_{zy}$. Both factors are integral; there are exactly 2^v inner labels. Scaling the row factor by c gives

$$\mathrm{VBound}(w, (cA)_{xr}), \qquad \mathrm{VBound}(w + v - u + 1, (J_{\mathrm{cross}} - cAB)_{xy}).$$

Feasibility $G_i \ge 0$ makes cA integral. At $v = a$ the factorization is exact and retains every original source row. PCC-20261114-9fb7c1/127309 [pcc:unverified].

Necessary lowest-layer count

Literal minor. Embed retained residues $r = 0, \dots, 2^u - 1$ as actual ports, valid because $u \le a$. The parent's constructed leading-layer proof implies that the resulting normalized minor M satisfies

$$\mathrm{VBound}(1, M_{rs} - [r = s]).$$

First derive entry integrality from this actual congruence. Finite-product congruence is proved by induction using the fixed ring rearrangement

$$ab - cd = (a - c)b + c(b - d).$$

Apply it termwise to the determinant's signed permutation sum. Integer signs are integral, so

$$\mathrm{VBound}(1, \det M - 1), \qquad \det M \neq 0, \qquad \mathrm{rank}_{\mathbb{Q}}\, M = 2^u.$$

This is the exact rank of this minor, not an exact rank claim for the whole rational matrix. PCC-20261114-4d783b [pcc:unverified].

Any approximate rational factors. Suppose rational matrices P, Q with k inner columns satisfy

$$\mathrm{VBound}(1, R_{xy} - (PQ)_{xy})$$

for every actual port pair. Restrict the product to the same literal ports. Its difference from identity has content one, hence its determinant is nonzero by the same lemma. Standard rational rank inequalities then give

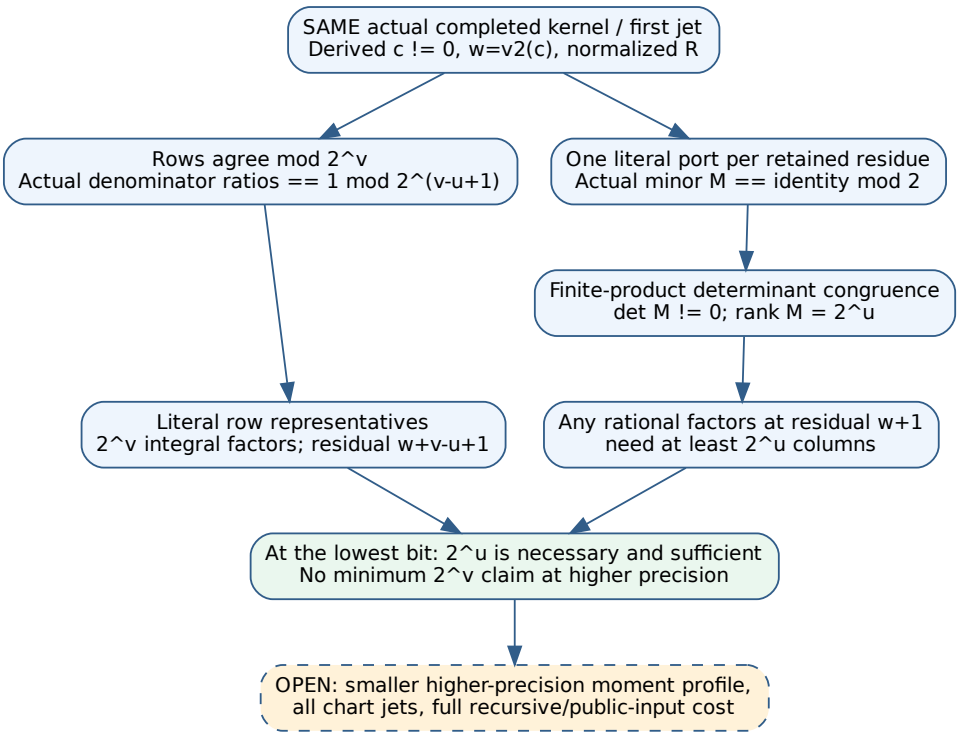
$$2^u = \mathrm{rank}(P_{\mathrm{res}} Q_{\mathrm{res}}) \le \mathrm{rank}(P_{\mathrm{res}}) \le k.$$

No integrality assumption on P or Q is needed. Dividing an actual unnormalized approximation of residual weight $w + 1$ by the derived nonzero c gives the same lower bound. PCC-20261114-1bce48/543d60 [pcc:unverified].

Precisely what matches. At $v = u$, the constructed upper control uses 2^u labels at normalized precision one; the lower theorem requires at least 2^u. At higher precision, the construction uses 2^v but no matching lower bound is asserted. For $k \ge 1$ and $u + k - 1 \le a$, the charged upper count is 2^{u+k-1}. Other moment constructions may use fewer labels.

These are bounds for the same particular kernel, first jet and approximation precision. They establish no lower bound for all completions, all algorithms or ECDLP. All higher chart jets and their actual weighted mode counts, coefficient construction, degree, recovery, entry and recursion costs remain required. Additive weights remain additive.

Evidence, scope and next proof obligation



Independent-check artifacts. Two modules contain40 authored statements. The pinned shared gate audits44 declarations, including4 generated congruence lemmas, with0 extra trusted axioms, exit0. Import-inclusive fresh OrientationRank is running at this checkpoint. All source-bound records are unsigned; gate receipt PCC-20261114-341743 [pcc:unverified].

Literal native artifact. Certificate IDC1hbc3887c49b3c9ce8 checks only $ab-cd=(a-c)b+c(b-d)$ on32+16 field points. It certifies neither parameter binding, valuation, determinant, rank nor cost. Three new adapter library tests, format and strict Clippy pass. No cryptographic fixture, performance/scaling experiment, paid/cloud/native search or old checker test was run [pcc:unverified].

Reproducibility and prior work. Exact signatures, hashes, gate log, parent pins, failed drafts and raw logs are retained. KB unavailable; read primary orientation/repeated-anchor/LocalCharts sources and pinned Mathlib determinant/rank proofs. Standard arithmetic and linear algebra are reused, with no priority claim. Protected concurrent ledger histories remain intact; PR4118 cannot merge before legitimate reconciliation and current-head gates.

Next discriminating work. Construct and charge the actual reciprocal- series moments within each retained residue class, comparing with the prior source's existing linear moments. Then cover all chart jets and the original recursive interface, with public-input finder/extraction and complete expected bit cost. First-jet bounds alone do not discharge those obligations.

`successor/PROOF_MAP.md`
`successor/evidence/claim-pins.json`
`successor/evidence/gate-validation-001.json`